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	Type I insulin dependent diabetes mellitus remains a major cause of morbidity and mortality. Exogenous insulin therapy only delays the onset and severity of end organ complications. While transplantation of whole organ pancreas can often cure diabetes, it is usually performed in diabetics who have developed kidney pathology since pancreas transplant is usually considered too serious a surgical risk as treatment for a disease treated with insulin. However, transplantation of pancreatic islets is a much less invasive surgical procedure and, when successful, it may help prevent complications. In addition, pre-transplant islet culture improves the survival of islets by reduction of immunogenicity. One problem with standard stationary islet culture, however, is a progressive loss of islets and their function. These problems may be avoided by use of a rotating wall vessel (RWV) bioreactor developed by NASA to culture cells while in continuous free fall. The bioreactor is characterized by high mass transfer of nutrients with significantly reduced turbulence, permitting		

Task Description:

cells to interact and form 3-dimensional (3-D) tissue-like configurations and in some instances to differentiate.
Our recent work shows clearly that mouse islets cultured in bioreactors have well preserved ultrastructural morphology and function, similar to fresh islets. Moreover, arrays of bioreactor-cultured islets develop intercellular channels with microvilli, which open onto the islet surface and may function to improve nutrition and facilitate graft vascularization. In contrast, stationary dish-cultured islets are less well preserved with signs of ultrastructural degeneration and they do not form large intercellular channels. Human islets suitable for transplantation are a scarce national resource. One donor pancreas can yield approximately 400,000-600,000 islets, which is below the number of islets necessary for successful allotranplantation thereby requiring more than one donor per recipient. To overcome these problems, proliferation and expansion of cultured human pancreatic adult stem cell populations and xenotransplantation of pig islets into Type 1 diabetics are promising approaches to address chronic shortages of donor islets. Xenotransplantation of pig islets is attractive because pigs are physiologically close to humans and represent a potentially unlimited islet resource for treatment of Type I diabetics. Cloning of pigs, offers the possibly of eliminating the threat of cross-species transmission of porcine endogenous retrovirus (PERV). The present research proposes two avenues of islet xenotransplantation in rat:mouse and pig:mouse models, namely: 1) to reduce immunogenicity and improve preservation of morphology and function by culturing rat and pig islets in bioreactors prior to transplantation, 2) to culture rat and pig islets in bioreactors with recipient mouse microvascular endothelial cells to seed large channels in islet grafts with vascular endothelium. In addition, this research proposes to expand and to differentiate in bioreactors populations of insulin-producing cell arrays isolated from rat, pig, or human non-islet pancreas tissues, as recently demonstrated in monolayer culture. We believe that microgravity culture conditions proffers a unique strategy for islet xenotransplantation and for development of insulin-producing cell arrays. These strategies should produce dramatic improvement in engraftment of xenogeneic pig islets and provide a secondary source of insulin-producing tissues that could be used as an adjunct for transplantation, thereby overcoming a critical shortage of human islet The overall objective of the proposed research is to evaluate the efficacy of culture of pancreatic islets in ground-based Rotating Wall Vessel (RWV) bioreactors for modulation of immunogenicity for xenotransplantation, differentiation of adult pancreatic stem cells into insulin-secreting cells, and enhancement of islet grafts vascularization by co-culture with microvascular endothelial cells. We plan to exploit the unique biological effects of modeled microgravity culture in bioreactors to produce: 1) physiologically superior rat and pig islets, with reduced immunogenicity and 2) to stimulate adult islet stem cells to differentiate into insulin-secreting cell arrays. Three hypotheses are to be tested: Hypothesis 1. Microgravity bioreactor culture condition reduces immunogenicity of rat or pig islets but preserves their morphology, making them a superior islet source for xenotransplantation in diabetic mice; Hypothesis 2. Co-culture of rat or pig islets with recipient mouse microvascular endothelial cells (EC) will seed the intercellular channels, which form only during bioreactor culture, promoting vascularization, and enhancing engraftment; and Hypothesis 3. Microgravity culture of rat, pig, and human non-islet pancreatic tissues, enhances ductal cell proliferation and differentiation into insulin-producing cell aggregates. The RWV bioreactor provides an opportunity to culture rat and pig islets for xenotransplantation in an environment that lowers immunogenicity and maintains excellent morphology and function. Moreover, this research investigates the effect of microgravity culture on proliferation and differentation of rat, pig and human pancreatic ductal stem cells into insulin-producing cell arrays. Finally, co-culture of microvascular endothelial cells with islets in the bioreactor environment may facilitate early engraftment and vascularization following transplantation. We believe that this technology may be applied to clinical islet transplantation and help ameliorate the critical shortage of pancreatic islets.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

During the past year, rat islet xenotransplantation studies were extended to include the Balb/c genetic strain of diabetic mouse recipients. Previously, diabetic C3H mice were used as recipients for rat islets. This new xenotransplant model will be used to study the impact of Stat signaling on prolonged xenograft survival in diabetic STAT4-/- and STAT6-/- knockout BALB/c mice (see below). In preliminary studies, 500 7-day bioreactor-cultured rat islets had prolonged survival, whereas, freshly isolated islets were acutely rejected. These results are similar to our earlier observations for rat xenografts transplanted in diabetic C3H mouse recipients. During the past year additional animals in each xenotransplant group for fresh rat islet, bioreactor-cultured islets, and dish-cultured islets. When compared with fresh islet xenotransplants, both bioreactor- and dish-culture significantly prolonged xenograft survival. Recently. xenotransplant studies were expanded to include human islets. For clinical islet transplantation, the gold standard for islet quality and potency prior to clinical transplantation is a xenograft bioassay in diabetic immunocompromised mice. In these studies, we evaluated marginal islet mass xenotransplants of 3-day bioreactor-cultured islets. Islets were transplanted under the kidney capsule in diabetic SCID mice and blood glucose levels and weight were monitored at intervals.
Bioreactor-cultured islets stained intensely red with dithizone, an insulin-binding dye, indicating the presence of high insulin content. When 1200 IE were transplanted, euglycemia was achieved within one day following transplant and was maintained for 39 days when an in vivo glucose tolerance test performed, which showed excellent islet function, indicating that a marginal islet mass of bioreactor-cultured human islets was potent and capable of reversing diabetes.
The impact of bioreactor culture on apoptosis (programmed cell death) in rodent and human islets was investigated during the past year. While islet transplantation with the newer steroid-free, reduced immunosuppression regimens offers great clinical promise in treating type 1 diabetes patients who are not well managed by intensive insulin therapy, pancreatic islets are sensitive to apoptosis, which probably accounts for considerable loss of functional islet mass during procurement, isolation and culture, and engraftment. In the current studies, human or rodent islets were placed either in stationary Petri dishes or bioreactor culture. In some studies a perfusion bioreactor system (Synthecon, Houston, TX) was used. As we demonstrated previously (Rutzky et al., 2002), bioreactor culture improved islet ultrastructure and hormone content. Apoptosis, evaluated by whole mount TUNEL staining for damaged DNA and evaluated by light microscopy and by silver enhanced nanogold TUNEL staining for electron microscopy, generally showed that bioreactor-culture reduced the amount of apoptosis, as evaluated on days 0, 1, 2, 4, and 6 and compared with stationary cultures.
Moreover, in islet aggregates apoptotic cells were often located near the periphery and between adjacent islets, which may be related to physical damage from collisions of islets during isolation and dish culture. Ultrastructural analysis confirmed that the majority of apoptotic cells were located at the periphery of the islets. Human islets cultured in a perfusion bioreactor had large nutritional channels between adjacent cells, which appeared similar to those observed in mouse islets bioreactor-cultures previously, in mouse islets bioreactor-cultures. These channels may help increase penetration of nutrients and oxygen into the islet central core, preserving structure and function and reducing apoptosis. These results indicate that culture of human islets in the bioreactor system may be helpful during the post-isolation period by improving islet quality prior to transplantation, allowing the recipient to begin pre-transplant immunosuppression treatment.

Task Progress:	We have continued to investigate the utility of bioreactor culturing to facilitate vascularization of islets prior to transplantation to enhance engraftment. In these preliminary studies, human islets were co-cultured with endothelial cells (HUVEC) human islets in bioreactors for 4 days. The composite-tissue islets were immunostained with two endothelial cell markers, Factor VIII and CD31 (PECAM) antibodies, to identify the presence of endothelial cells within islets. These studies are designed to test the hypothesis that channel formation during bioreactor culture of islets that may function as a scaffold for growth of recipient endothelial cells prior to transplantation. In these studies, human pancreas tissue controls had some areas of cell-associated anti-CD31 (PECAM) immunoperoxidase staining, with stained cells appearing to be associated with small vessel-like structures located throughout the islets. Pancreas tissue also had limited Factor VIII staining. Monolayer-cultured HUVEC cells stained deeply with both CD31 and Factor VIII antibodies. Human islets that were shipped to our laboratory had CD31 positive cells located on the islet periphery and inside the islets, with fewer Factor VIII positive cells. Bioreactor-cultured islets co-cultured with HUVEC cells had specific cell-associated staining for both Factor VIII and CD31 located inside the islets. Bioreactor cultured, HUVEC-free control islets had no immunostaining cells. Stationary dish cultures had very limited, background CD31 and Factor VIII staining. These results suggest that bioreactor culturing may facilitate entrance and possible growth of HUVEC endothelial-cells via the surface pores and interior channels, providing preliminary evidence that the bioreactor system may be used to seed islets with recipient endothelial cells with enhanced engraftment. To test the hypothesis that recipient endothelial cells will facilitate islet engraftment, we will next prepare composite tissues of mouse endothelial cell strain, MLEC, and rat islets in the bioreactor, which will be transplanted to diabetic syngeneic mouse recipients. During the past year, we investigated whether immune regulation occurs in our bioreactor-cultured mouse islet model system with long-term allograft survival (mean survival time, MST>100 days), without any exogenous immunosuppression. To test this hypothesis, 3-day bioreactor cultured C57BL10 mouse donor islet allografts were transplanted under the kidney capsule of diabetic Signal Transducer and Activators of Transcription (STAT), STAT4 -/-and STAT6 -/- knockout BALB/c mice. STAT 4-/- is associated with losses in Th1 cell function, but retains STAT6 dependent immune regulation, protecting long-term allograft survival (tolerance). Whereas, STAT 6-/- mice have losses in Th2 cell function, possess no STAT 6-associated immune regulation, and retain potent STAT4-dependent Th1 rejection of islet allografts. STAT4-dependent signaling promotes generation of interleukin (IL)-2 and interferon (IFN)-producing T helper 1 (Th1) lymphocytes, whereas, STAT6-dependent signaling promotes IL-4/IL-10-producing Th2 cells. In normal animals, IL-12/STAT4-associated signaling in Th1 cells promotes allograft rejection and IL-4/STAT6-signaling in Th2 cells promotes generation of transplantation tolerance. In these studies, C57Bl/10 (H2b) islets were cultured in disposable RWV bioreactors (bioreactor) for 3 and 7 days prior to transplantation (500 islets under the kidney capsule) in diabetic Balb/c mice (H2d), a major MHC mismatch strain combination. As we have shown in the past, freshly isolated mouse islet allografts were acutely rejected at approximately 13 days. When islets were cultured in bioreactors for 7 days, long-term allograft survival (>100 days) occurred, confirming our previous observations (Rutzky et al., 2002). Islets cultured for 3 days had prolonged graft survival, but were eventually rejected. The difference in MST between 3 and 7day bioreactor cultured islets and the difference in MST between 7-day culture and freshly isolated islets were statistically significant. Based on these observations that 3-day cultured islets are rejected, but 7 day-cultured islet allografts are accepted long-term, we postulate that in addition to depletion of antigen presenting and MHC Class II cells, culture of islets may also be associated with immune regulation. To test this hypothesis, mouse islet allografts were performed in diabetic STAT4-/- and STAT6-/- knockout BALB/c mice recipients. Specifically, fresh C57Bl/10 islets transplanted under the kidney capsule of diabetic STAT4-/- recipients were acutely rejected, whereas, after 3 days of culture in the bioreactor, long-term graft survival was observed. Similarly, fresh islets transplanted into diabetic STAT6-/- recipients, were also acutely rejected. In contrast, in Stat6-/- recipients with islet allografts cultured for 3 days in the bioreactor showed a mixed response with some animals maintaining long-term grafts, while, some animals rejected their grafts in a delayed fashion. These results suggest that Stat6-dependent regulation is involved in maintenance of long-term surviving islet allografts and are similar to results published by Kishimoto et al. (J. Clin. Invest. 106: 63, 2000) in Stat6-/- and Stat4-/- recipients of heart allografts for tolerance induced by CD40L and CTLA4-Ig co-stimulatory blockade. To investigate whether immune regulation is present in long-term surviving allografts of bioreactor-cultured islets, an adoptive transfer model system is currently being evaluated. These studies are designed to investigate whether culturing islets in bioreactors prior to allograft transplantation can facilitate the development of regulatory T cells and the induction of immune tolerance.
Bibliography Type:	Description: (Last Updated: 06/08/2005)
Abstracts for Journals and Proceedings	L. Rutzky, M. Kloc, S. Bilinski, H. Zhang, T. Phan, S. Katz, and S. Stepkowski "Modeled Microgravity Culture of Human Pancreatic Islets Improves Function and Reduces Apoptosis." In Vitro Cellular & Developmental Biology 40: 37-A. 2004 , May-2004